Survival Analysis

Mid-term project

STDP-based spiking deep convolutional neural networks for object recognition

Summary

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Reference :

STDP-based spiking deep convolutional neural networks for object recognition. Neural Networks (2017)

Kheradpisheh, S.R., Ganjtabesh, M., Thorpe, S.J., Masquelier, T

https://arxiv.org/pdf/1611.01421.pdf

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Deep Convolutional Neural Networks (DCNNs) have been proven to be high performing and able to learn complex features, especially when it comes to object recognition. However, DCNNs don't include the biological neural process which takes place in the human cortex. They are slow to converge and need millions of labelled examples in order to avoid overfitting, while the human brain learns fast with few examples.
This problem led to the birth of the 3rd generation of Neural Networks (NNs) which aims to bridge the gap between neuroscience and machine learning: Spiking Neural Networks (SNNs). SNNs operate using spikes, which are discrete events in time, rather than continuous values. Essentially, once a neuron reaches a certain potential, it spikes, and the potential of that neuron is reset. Nevertheless, SNNs are still slow and based on learning methods which are not biologically plausible. As a trial to solve this problem, Spike Timing Dependent Plasticity (STDP) can be used in SNNs. STDP is an unsupervised learning mechanism which occurs in mammalian visual cortex. It adjusts the strength of connections between neurons in the brain based on the timing of a particular neuron's output and input action potential (or spikes): Synapses[1] through which a presynaptic spike arrived before (respectively after) a postsynaptic one are reinforced (respectively depressed). STDP has been proven to be applicable in SNNs to detect visual features without supervised learning. However, all previous studies on STDP-based SNNs used a simplified structure based on only one trainable layer (low complexity models).
This paper tries to bridge the gap between deep learning, machine learning, and neuroscience, by introducing a STDP-based Spiking **Deep** Neural Network (SDNN).

Suggested model

In order to recognise objects, STDP-based SDNNs try to imitate the human primate visual system which proceeds in a hierarchical feed-forward way. Multiple layers are used (deep learning), and neurons in each layer learn how to detect frequent features in an unsupervised way and with an increasing complexity of visual features, reaching object prototypes detection in the last layer.

Model's architecture

The model contains three types of layers and a classifier at the end. A sample architecture of the proposed SDNN, with 3 convolutional layers and 3 pooling layers, is presented in figure 1 below.

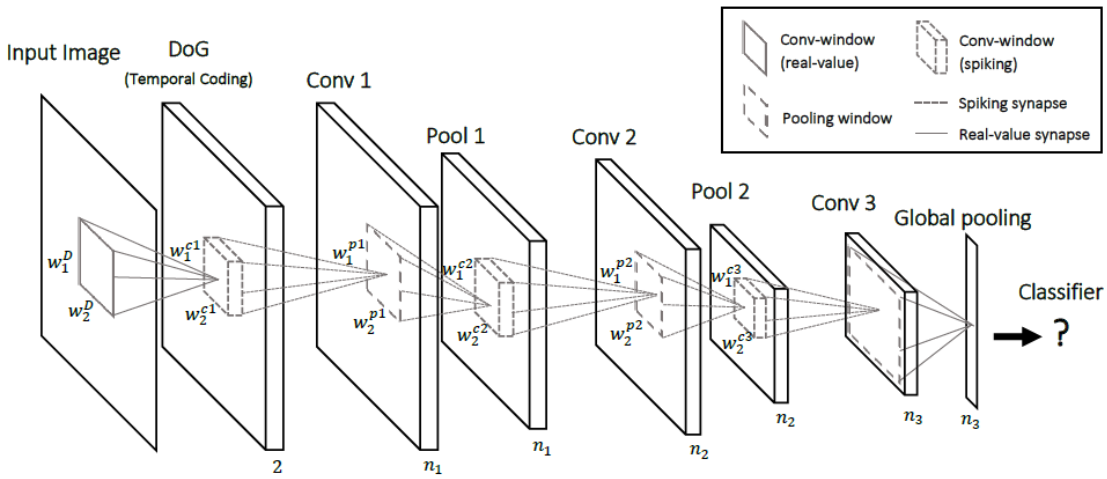

Figure 1: Simplified architecture of the proposed SDNN

[1]In the nervous system, a synapse is a structure that permits a neuron (or nerve cell) to pass an electrical or chemical signal to another neuron or to the target effector cell.

Temporal encoding layer: This layer converts the input image into an asynchronous spike train, where the visual information is encoded in the temporal order of the spikes. Cells in this layer use Differences of Gaussians (DoG) filters[2] to detect contrasts in the input image. It encodes the strength of these contrasts in the latencies of its output spikes (the higher the cell's contrast, the earlier it fires). The temporal encoding scheme is important since it determines the information amount in each spike, hence affects the speed and accuracy. For faster computing, the input spikes are grouped and propagated in equal-size sequential packets.

Convolutional layers: Those layers are the learning layers. They integrate input spikes and emit a spike after reaching their threshold. Each convolutional layer contains several neuronal maps. Neurons in the same map detect the same specific visual feature (hence have the same weights), but in different locations. Different maps detect different visual features. The visual feature in each layer is a combination of simpler extracted features in the previous layer. In order to ensure a sparse yet highly informative encoding, neurons fire at most once in each location, and potentials of all neurons in that location are reset to zero if at least one of the neurons fires.

Pooling layers: A process of max-pooling occurs within these layers. It is done by applying a max-filter to subregions of the initial representation. For each of the regions represented by the filter, the max of that region is taken and a new output matrix, where each element is the max of a region in the original input, is created.

The objective of max-pooling is to downsample the input representation, reducing its dimensionality, hence compacting the visual information. Besides reducing the number of parameters to learn, max-pooling provides basic translation invariance to the internal representation (for example, an image should be classified the same way whether it is upside down or not). No learning occurs in these layers.

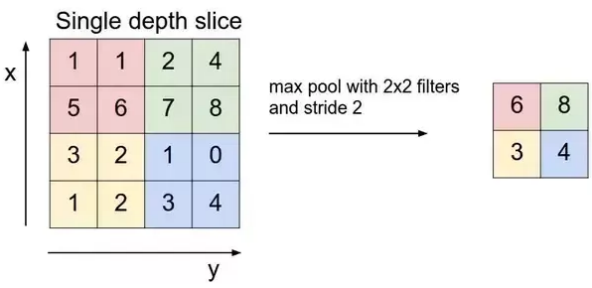

Figure 2: Max-pooling example

Learning and recognition process

After the object image is encoded in the temporal encoding layer, the output spikes are received by the convolutional layers. Learning occurs only in convolutional layers, and in a sequential way (learning in a layer starts after learning in the previous one is finished). A simplified version of the STDP learning process is as follows:

$$\Delta w_{i,j} = \begin{cases} a^{+} w_{i,j}(1 - w_{i,j}) & \text{if } t_j - t_i \leq 0 \\ a^{-} w_{i,j}(1 - w_{i,j}) & \text{if } t_j - t_i > 0 \end{cases}$$

Where i and j respectively refer to the index of post and presynaptic neurons, t_i and t_j are the corresponding spike times, Δw_{ij} is the synaptic weight modification, and a^+ and a^- are two parameters specifying the learning rate. Only the sign of time difference affects the weights change. The learning parameters (i.e. a^+ and a^-) should be chosen wisely; Large values decrease the learning memory (neurons will remember only recent images), while very small values slow down the learning process.

As mentioned before, neurons in the same map detect the same feature but in different locations. They compete with each other. The first one to reach the threshold and fire, is the winner which

[2]DoG is a feature enhancement algorithm that involves the subtraction of one blurred version of an original image from another, less blurred version of the original. It is a band-pass filter that discards all but a handful of spatial frequencies that are present in the original image.

triggers STDP, learns the feature, then updates its weights. Since neurons belonging to the same map should have the same weights, the winner updates its weights to the rest. In order to make different maps learn different features, the winner prevents other neurons from other maps to do the STDP, and that within a small neighbourhood.

After the learning phase in each convolutional layer comes the pooling phase in the corresponding pooling layer. We remind that this phase allows translation invariance and information compaction. The output of the final pooling layer is used to train a linear SVM classifier. The classifier is then used in the testing phase for image category recognition.

Experiments and results

The proposed SDNN was evaluated on Caltech face/motorbike and ETH-80 datasets with large scale images of various objects taken form different viewpoints. The proposed SDNN reached the accuracies of 99.1% on face/motorbike task and 82.8 % on ETH-80, which indicates its capability to recognise several natural objects even under severe variations. The proposed SDNN was also evaluated on the MNIST data set which is a benchmark for spiking neural networks, and it reached 98.4 % recognition accuracy. In addition to the high performance compared to other techniques (figures 3 and 4 below), the proposed SDNN is highly energy-efficient and works with a few number of spikes per image, which makes it suitable for neuromorphic hardware implementation.

Method	Accuracy (%)
HMAX [23]	69.0
Convolutional SNN [23]	81.1
Pre-trained AlexNet	79.5
Fine-tuned AlexNet	94.2
Supervised DCNN	81.9
Unsupervised DCA	80.7
Proposed SDNN	82.8

Figure 3: Results over ETH-80 data set

Architecture	Neural coding	Learning-type	Learning-rule	Accuracy (%)
Dendritic neurons [21]	Rate-based	Supervised	Morphology learning	90.3
Convolutional SNN [59]	Spike-based	Supervised	Tempotron rule	91.3
Two layer network [44]	Spike-based	Unsupervised	STDP	93.5
Spiking RBM [39]	Rate-based	Supervised	Contrastive divergence	94.1
Two layer network [11]	Spike-based	Unsupervised	STDP	95.0
Convolutional SNN [12]	Rate-based	Supervised	Back-propagation	**99.1**
Proposed SDNN	Spike-based	Unsupervised	STDP	98.4

Figure 4: Results over MNIST data set

Insights

Although current state-of-the-art DCNNs achieved stunning results on various recognition tasks, continued work on brain-inspired models may help gain more insights into the learning process of the primate visual system, and the human brain in general. Combining STDP and latency coding can also be used in artificial intelligence systems, especially hardware solutions.